Milestone 1 (Draft)

CSC 676 - Spring 2026

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Version History:

Version	Description	Date
v0.1	Initial draft for in-class review; sample entities and requirements.	2026-02-08
v0.2	Update for draft #2; Functional Database complete, conceptual design begun, Entity Set Desc. Begun	2026-02-17

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Project Description

- **High-Level Overview:**
    - The Transit Incident & Maintenance Reporting System improves how passengers experience and interact with local rail platforms like BART and Muni. Unlike existing apps that focus on static schedules and basic navigation, our platform delivers live, actionable data: real-time train crowding, station congestion, accurate next-train predictions, and immediate incident reporting for accidents or hazards. With this system, passengers get the information they need to make smarter travel decisions, while transit operators and first responders gain a unified view of system-wide conditions to dispatch resources more effectively. A public transparency portal keeps riders informed of active issues and resolutions, reducing duplicate reports and building trust in the system.

- **Unique Features:**
    - Recursive Delay Propagation: Use recursive relationships to automatically calculate the “downstream” impact of a single train delay on every station following it on the line.
    - Community Transparency Portal: To be more efficient than existing platforms like the “BART Watch” App. To improve the passenger experience, our system includes a public-facing communication space. By having this, it is easier for the community to see follow-up notes on active messes or delays. It prevents the issue of duplicate reporting by allowing users to see if an issue is already 'In-Progress' or 'Resolved' in real-time.

Functional Database Requirements

1. **User**
    - 1.1. A user shall have exactly one unique user ID.
    - 1.2. A user shall have exactly one unique username.
    - 1.3. A user shall have exactly one password.
    - 1.4. A user shall submit zero or many reports.

2. **Admin**
    - 2.1. An admin shall reference exactly one user ID.
    - 2.2. An admin shall have exactly one unique admin ID.
    - 2.3. An admin shall manage system data, including stations, lines, and agencies.

3. **Employee**
    - 3.1. An employee shall reference exactly one user ID.
    - 3.2. An employee shall have exactly one unique employee ID.
    - 3.3. An employee shall be assigned to exactly one station.

4. **Officer**
    - 4.1. An officer shall reference exactly one user ID.
    - 4.2. An officer shall have exactly one unique officer ID.
    - 4.3. An officer shall have exactly one badge number.

5. **Agency**
    - 5.1. An agency shall have exactly one unique agency ID.
    - 5.2. An agency shall have exactly one unique name.
    - 5.3. An agency shall operate zero or many lines.
    - 5.4. An agency shall operate zero or more vehicles.
    - 5.5. An agency shall serve zero or many stations through Station_Agency.

6. **Station**
    - 6.1. A station shall have exactly one unique station ID.
    - 6.2. A station shall have exactly one name.

6.3. A station shall be served by many agencies through Station_Agency.

6.4. A station shall belong to zero or many lines.

6.5. A station shall be assigned to zero or more employees.

6.6. A station shall be associated with zero or many lines through Line_Station.

7. Station_Agency

7.1. A Station_Agency shall reference exactly one station ID.

7.2. A Station_Agency shall reference exactly one agency ID.

7.3. A Station_Agency (station ID, agency ID) pair shall be unique.

8. Mode (of transport)

8.1. A mode shall have exactly one unique mode ID.

8.2. A mode shall have exactly one unique name.

8.3. A mode shall be assigned to zero or many lines.

9. Line

9.1. A line shall have exactly one unique line ID.

9.2. A line shall have exactly one unique name.

9.3. A line shall belong to exactly one agency.

9.4. A line shall be assigned to exactly one mode.

9.5. A line shall be associated with zero or many stations through Line_Station.

9.6. A line shall be assigned to zero or many trips.

10. Line_Station

10.1. A Line_Station shall reference exactly one line ID.

10.2. A Line_Station shall reference exactly one station ID.

10.3. A Line_Station shall store exactly one sequence number.

10.4. A Line_Station (line ID, station ID) pair shall be unique.

10.5. A Line_Station (line ID, sequence number) pair shall be unique.

11. Incident_Report

11.1. An Incident_Report shall have exactly one unique incident ID.

11.2. An Incident_Report shall be reported by exactly one user.

11.3. An Incident_Report shall be assigned to zero or many employees through Dispatch_Assignment.
11.4. An Incident_Report shall have zero or many Incident_Photo.
11.5. An Incident_Report shall belong to zero or many stations through Incident_Station.
11.6. An Incident_Report shall belong to zero or many trains through Incident_Train.
11.7. An incident_Report shall have exactly one description.
11.8. An Incident_Report shall reference exactly zero or one Incident_Status.
11.9. An Incident_Report shall reference exactly one Report_Category.
11.10. An Incident_Report shall reference exactly one weather status.
11.11. An Incident_Report shall reference zero or many hazards.

12. Incident_Station

12.1. An Incident_Station shall reference exactly one Incident_Report ID.
12.2. An Incident_Station shall reference exactly one Station ID.
12.3. A Incident_Station (Incident_Report ID, station ID) pair shall be unique

13. Incident_Train

13.1. An Incident_Train shall reference exactly one incident report ID.
13.2. An Incident_Train shall reference exactly one train ID.
13.3. An Incident_Train (incident report ID, train ID) pair shall be unique.

14. Incident_Photo (Weak)

14.1. An Incident_Photo shall have exactly one unique Incident_Photo ID.
14.2. An incident_Photo shall be attached to exactly one Incident_Report.
14.3. An Incident_Photo shall have exactly one timestamp.

15. Incident_status

15.1. An Incident_Status shall have exactly one unique status ID.
15.2. An Incident Status shall have exactly one unique name.
15.3. An Incident_Status shall be assigned to zero or many Incident_Report

16. Incident_History

16.1. An Incident_History shall have exactly one unique history ID.

16.2. An Incident_History shall reference exactly one incident report ID.

16.3. An Incident_History shall store exactly one timestamp.

16.4. An Incident_History shall store exactly one action description.

17. Report_Category (Safety, Maintenance, Environmental)

17.1. A Report_Category shall have exactly one unique category ID.

17.2. A Report_Category shall have exactly one unique category name.

17.3. A Report_category shall be assigned to zero or many Incident_Report.

18. Dispatch_Low_Level

18.1. A Dispatch_Low_Level shall have exactly one unique low level dispatch ID.

18.2. A Dispatch_Low_Level shall reference exactly one incident report ID.

18.3. A Dispatch_Low_Level shall reference zero or more employee ID (or zero).

18.4. A Dispatch_Low_Level shall reference zero or more officer ID (or zero).

18.5. A Dispatch_Low_Level shall have at least one of employee ID or officer ID.

19. Car

19.1. A car shall have exactly one unique car ID.

19.2. A car shall have exactly one unique car name.

19.3. A vehicle shall belong to exactly one agency.

19.4. A car shall have exactly one car capacity.

19.5. A car shall reference exactly one Vehicle_Status.

20. Train

20.1. A train shall have exactly one unique train ID.

20.2. A train shall belong to exactly one agency.

20.3. A train shall consist of one or many cars through Train_Car.

20.4. A train shall reference exactly one Vehicle_Status.

21. Train_Car

21.1. A Train_Car shall reference exactly one Train ID.

21.2. A Train_Car shall reference exactly one car ID.

21.3. A Train_Car (train ID, car ID) pair shall be unique.

22. Vehicle_Status

22.1. A vehicle status shall have exactly one unique status ID.

22.2. A vehicle status shall have exactly one unique status name (e.g., 'Active', 'Maintenance', 'Out of Service').

22.3. A vehicle status shall be assigned to zero or many cars.

22.4. A vehicle status shall be assigned to zero or many trains.

23. Distance

23.1. A distance shall have exactly one unique distance ID.

23.2. A distance shall reference exactly one user ID.

23.3. A distance shall reference exactly one station ID.

23.4. A distance shall store exactly one value in miles.

23.5. A distance shall have exactly one calculated timestamp.

23.6. A (user ID, station ID) pair may appear multiple times (different timestamps).

24. Weather

24.1. A weather status shall have exactly one unique weather ID.

24.2. A weather status shall store exactly one weather status (Sourced from local meteorology networks).

24.3. A weather status shall reference exactly one data feed ID.

24.4. A weather shall be assigned to zero or many incident_reports to indicate weather at the time of the incident.

25. Hazards

25.1. A hazard shall have exactly one unique hazard ID.

25.2. A hazard shall have exactly one unique name.

25.3. A hazard shall be assigned to zero or many Incident_Report to indicate associated hazards.

26. Incident_Hazard

26.1. An Incident_Hazard shall reference exactly one incident report ID.

26.2. An Incident_Hazard shall reference exactly one hazard ID.

26.3. An Incident_Hazard (incident report ID, hazard ID) pair shall be unique.

27. Priority_Level

27.1. A Priority_Level shall have exactly one unique priority ID.

27.2. A Priority_Level shall have exactly one priority name.

27.3. A Priority_Level shall be assigned to zero or many incident reports.

28. Saved_Station

28.1. A Saved_Station shall have exactly one unique saved station ID.

28.2. A Saved_Station shall reference exactly one user ID.

28.3. A Saved_Station shall reference exactly one station ID.

28.4. A Saved_Station shall have exactly one created timestamp.

28.5. A user shall have zero or many saved stations.

28.6. A (user ID, station ID) pair shall be unique.

29. Upvote

29.1. An upvote shall have exactly one unique upvote ID.

29.2. An upvote shall be created by exactly one user.

29.3. An upvote shall reference exactly one incident report.

29.4. An upvote shall have exactly one created timestamp.

29.5. A user shall create zero or many upvotes.

29.6. An incident report shall receive zero or many upvotes.

29.7. A (user ID, incident report ID) pair shall be unique.

30. Notification_Preference

30.1. A notification preference shall have exactly one unique preference ID.

30.2. A notification preference shall reference exactly one user.

30.3. A notification preference shall have exactly one notification type.

30.4. A notification preference shall have exactly one enabled flag.

30.5. A user shall have zero or many notification preferences.

30.6. A (user ID, notification type) pair shall be unique.

31. Notification

31.1. A notification shall have exactly one unique notification ID.

31.2. A notification shall reference exactly one user.

31.3. A notification shall reference zero or one incident report.

31.4. A notification shall reference zero or one line.

31.5. A notification shall have exactly one created timestamp.

31.6. A notification shall have exactly one delivery status.

31.7. A user shall receive zero or many notifications.

31.8. A notification shall reference exactly one notification type.

32. Trip

32.1. A Trip shall have exactly one unique trip ID.

32.2. A Trip shall be operated by exactly one train.

32.3. A Trip shall follow exactly one line.

32.4. A Trip shall have exactly one scheduled start time.

32.5. A Trip shall have exactly one scheduled end time.

33. Stop

33.1. A Stop shall have exactly one unique stop ID.

33.2. A Stop shall reference exactly one trip ID.

33.3. A Stop shall reference exactly one station ID.

33.4. A Stop shall have exactly one scheduled arrival time.

33.5. A Stop shall have exactly one scheduled departure time.

33.6. A Stop shall have exactly one stop sequence number.

33.7. A (trip ID, station ID) pair shall be unique.

33.8. A (trip ID, stop sequence number) pair shall be unique.

34. Live_Feed

34.1. A Live_Feed shall have exactly one unique feed ID.

34.2. A Live_Feed shall reference exactly one stop ID.

34.3. A Live_Feed shall have exactly one update timestamp.

34.4. A Live_Feed shall store exactly one estimated arrival time.

34.5. A Live_Feed shall store exactly one current train capacity (percentage).

35. Delay_Propagation

35.1. A Delay_Propagation shall have exactly one unique propagation ID.

35.2. A Delay_Propagation shall reference exactly one incident report ID.

35.3. A Delay_Propagation shall reference exactly one trip ID.

35.4. A Delay_Propagation shall reference exactly one stop ID.

35.5. A Delay_Propagation shall store exactly one calculated delay duration (in minutes).

36. Update

36.1. An Update shall have exactly one unique update ID.

36.2. An Update shall reference exactly one incident report ID.

36.3. An Update shall be created by exactly one user.

36.4. An Update shall have exactly one comment text.

36.5. An Update shall have exactly one created timestamp.

37. Vote

37.1. A Vote shall have exactly one unique vote ID.

37.2. A Vote shall be created by exactly one user.

37.3. A Vote shall reference exactly one update ID.

37.4. A Vote shall have exactly one vote type (e.g., 'Confirm' or 'Helpful').

37.5. A (user ID, update ID) pair shall be unique.

38. Responder

38.1. A Responder shall have exactly one unique responder ID.

38.2. A Responder shall have exactly one unique responder name (e.g., 'Police', 'Fire', 'EMS').

38.3. A Responder shall have exactly one description (e.g., 'Law enforcement unit', 'Fire suppression and rescue', 'Emergency medical services').

39. Dispatch_High_Level

39.1. A Dispatch_High_Level shall have exactly one unique high level dispatch ID.

39.2. A Dispatch_High_Level shall reference exactly one incident report ID.

39.3. A Dispatch_High_Level shall reference exactly more than one responder ID.

39.4. A Dispatch_High_Level shall reference exactly one station ID.

39.5. A Dispatch_High_Level shall store exactly one dispatched timestamp.

40. Fare_Log

40.1. A Fare_Log shall have exactly one unique log ID.

40.2. A Fare_Log shall reference exactly one incident report ID.

40.3. A Fare_Log shall reference exactly one user ID.

40.4. A Fare_Log shall reference exactly one station ID.

40.5. A Fare_Log shall have exactly one detected timestamp.

41. Data_Feed

41.1. A Data_Feed shall have exactly one unique feed ID.

41.2. A Data_Feed shall have exactly one unique feed name (e.g., 'National Weather Service').

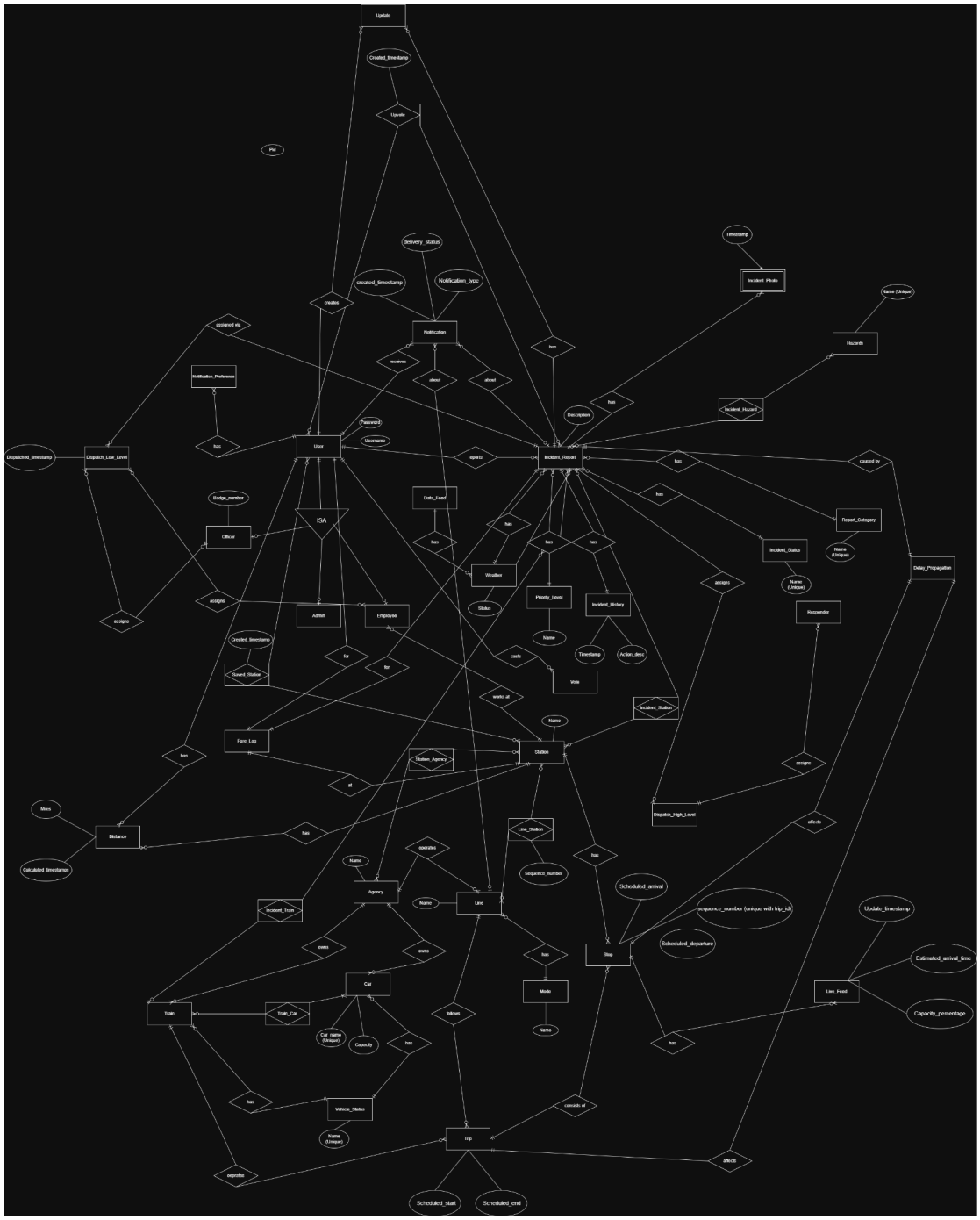
41.3. A Data_Feed shall have exactly one feed type (e.g., 'Weather', 'Schedule').

Non-Functional Requirements

1. **Security & Privacy**
    - 1.1. Publicly visible reports shall hide the name of the reporter.
    - 1.2. “Criminal” reports shall be reviewed by staff until approved for the public feed.
    - 1.3. User passwords shall be encrypted in the database.
2. **Performance**
    - 2.1. Community portal shall support up to 5,000 concurrent users.
    - 2.2. Delay calculations shall complete within a few seconds.
    - 2.3. The system shall support 500 concurrent users during peak hours.
3. **Scalability**
    - 3.1. BART and Muni services are covered by our services.
    - 3.2. This database design will support adding new transit agencies without major changes in structure.
    - 3.3. The system shall handle 10,000 incidents without performance degradation.
4. **Reliability and Availability**
    - 4.1. Database backups should be performed daily.
    - 4.2. Database backups should be stored securely.
    - 4.3. The system shall have 100% uptime during operating hours.
5. **Storage**
    - 5.1. Incident reports shall be retained for at least 90 days.
    - 5.2. Each incident should support storage for up to 5 incident photos.
    - 5.3. Station and line data shall be pre-populated for BART and Muni.
6. **Privacy**
    - 6.1. Reporter name shall not appear in public view.
    - 6.2. User email addresses shall only be used only for notification.
    - 6.3. The database shall store only necessary user information.
7. **Maintainability and Coding Standards**
    - 7.1. Naming conventions shall be consistent across all tables.
    - 7.2. Each table should contain relevant comments describing its purpose.
    - 7.3. The database design shall be accompanied by an Enhanced Entity-Relationship Diagram.

- **Tradeoffs**:
    - By having public reports, the database will have more read requests and will be slower. However, we will limit report posting to one person at a time to prevent overflow of data.

Conceptual Design Enhanced Entity-Relationship Diagram



Entity Set Descriptions

1. **User**
   - 1.1. Primary Key(s): user_id
   - 1.2. Foreign Key(s): None
   - 1.3. Attribute(s): username (unique), password
   - 1.4. Description: Represents a system user who can submit incident reports and interact with the platform.

2. **Admin**
   - 2.1. Primary Key(s): admin_id
   - 2.2. Foreign Key(s): user_id (references User)
   - 2.3. Attribute(s): None
   - 2.4. Description: System administrators who manage transit data including stations, lines, and agencies.

3. **Employee**
   - 3.1. Primary Key(s): employee_id
   - 3.2. Foreign Key(s): user_id (references User), station_id (references Station)
   - 3.3. Attribute(s): None
   - 3.4. Description: Transit agency staff assigned to specific stations who can be dispatched to incidents.

4. **Officer**
   - 4.1. Primary Key(s): officer_id
   - 4.2. Foreign Key(s): user_id (references User)
   - 4.3. Attribute(s): badge_number (unique)
   - 4.4. Description: Transit police or security personnel with badge numbers who can be dispatched to incidents.

5. **Agency**
   - 5.1. Primary Key(s): agency_id
   - 5.2. Foreign Key(s): None

5.3. Attribute(s): name (unique)

5.4. Description: Transit operators that own and operate lines, vehicles, and serve stations.

6. Station

6.1. Primary Key(s): station_id

6.2. Foreign Key(s): None

6.3. Attribute(s): name

6.4. Description: Physical transit stops where passengers board and alight.

7. Station_Agency

8. Primary Key(s): (station_id, agency_id) composite

9. Foreign Key(s): station_id (references Station), agency_id (references Agency)

10. Attribute(s): None

11. Description: Links stations to the agencies that serve them.

12. Mode

12.1. Primary Key(s): mode_id

12.2. Foreign Key(s): None

12.3. Attribute(s): name (unique)

12.4. Description: Types of transportation that lines can use (e.g., light rail, subway).

13. Line

14. Primary Key(s): line_id

15. Foreign Key(s): agency_id (references Agency), mode_id (references Mode)

16. Attribute(s): name (unique)

17. Description: Named transit routes that follow a specific sequence of stations.

18. Line_Station

18.1. Primary Key(s): (line_id, station_id) composite

18.2. Foreign Key(s): line_id (references Line), station_id (references Station)

18.3. Attribute(s): sequence_number (unique with line_id)

18.4. Description: Defines the ordered sequence of stations on a line.

19. Incident_Report

19.1. Primary Key(s): incident_id

19.2. Foreign Key(s): user_id (references User), status_id (references Incident_Status, optional), category_id (references Report_Category), weather_id (references Weather), priority_id (references Priority_Level)

19.3. Attribute(s): description

19.4. Description: Core entity for all reported issues including accidents, hazards, maintenance needs, and safety concerns.

20. Incident_Station

20.1. Primary Key(s): (incident_id, station_id) composite

20.2. Foreign Key(s): incident_id (references Incident_Report), station_id (references Station)

20.3. Attribute(s): None

20.4. Description: Links incidents to the stations where they occurred.

21. Incident_Train

21.1. Primary Key(s): (incident_id, train_id) composite

21.2. Foreign Key(s): incident_id (references Incident_Report), train_id (references Train)

21.3. Attribute(s): None

21.4. Description: Links incidents to specific trains involved.

22. Incident_Photo

22.1. Primary Key(s): photo_id

22.2. Foreign Key(s): incident_id (references Incident_Report)

22.3. Attribute(s): timestamp

22.4. Description: Images uploaded by users to document incidents.

23. Incident_Status

23.1. Primary Key(s): status_id

23.2. Foreign Key(s): None

23.3. Attribute(s): name (unique)

23.4. Description: Lookup table for incident lifecycles (e.g., 'Reported', 'In Progress', 'Resolved').

24. Incident_History

24.1. Primary Key(s): history_id

24.2. Foreign Key(s): incident_id (references Incident_Report)

24.3. Attribute(s): timestamp, action_desc

24.4. Description: Audit log of all actions taken on an incident.

25. Report_Category

25.1. Primary Key(s): category_id

25.2. Foreign Key(s): None

25.3. Attribute(s): name (unique)

25.4. Description: Classification of incident types (Safety, Maintenance, Environmental).

26. Dispatch_Low_Level

26.1. Primary Key(s): low_level_dispatch_id

26.2. Foreign Key(s): incident_id (references Incident_Report), employee_id (references Employee, optional), officer_id (references Officer, optional)

26.3. Attribute(s): dispatched_timestamp

26.4. Description: Records which employees and officers are assigned to respond to incidents. At least one of employee or officer must be present.

27. Car

27.1. Primary Key(s): car_id

27.2. Foreign Key(s): agency_id (references Agency), vehicle_status_id (references Vehicle_Status)

27.3. Attribute(s): car_name (unique), capacity

27.4. Description: Individual railcars that can be coupled to form trains.

28. Train

28.1. Primary Key(s): train_id

28.2. Foreign Key(s): agency_id (references Agency), vehicle_status_id (references Vehicle_Status)

28.3. Attribute(s): None

28.4. Description: Assembled consists of one or more railcars that operate as a unit.

29. Train_Car

29.1. Primary Key(s): (train_id, car_id) composite

29.2. Foreign Key(s): train_id (references Train), car_id (references Car)

29.3. Attribute(s): None

29.4. Description: Links trains to their constituent cars over time.

30. Vehicle_Status

30.1. Primary Key(s): status_id

30.2. Foreign Key(s): None

30.3. Attribute(s): name (unique)

30.4. Description: Lookup table for vehicle operational status (Active, Maintenance, Out of Service).

31. Distance

31.1. Primary Key(s): distance_id

31.2. Foreign Key(s): user_id (references User), station_id (references Station)

31.3. Attribute(s): miles, calculated_timestamp

31.4. Description: Stores calculated distances between a user's location and a station at a specific time. Multiple records per user-station pair allowed.

32. Weather

32.1. Primary Key(s): weather_id

32.2. Foreign Key(s): data_feed_id (references Data_Feed)

32.3. Attribute(s): status

32.4. Description: Weather conditions at the time of incidents, sourced from external meteorological networks.

33. Hazards

33.1. Primary Key(s): hazard_id

33.2. Foreign Key(s): None

33.3. Attribute(s): name (unique)

33.4. Description: Lookup table for specific danger types (e.g., spill, fire, downed power line).

34. Incident_Hazard

34.1. Primary Key(s): (incident_id, hazard_id) composite

34.2. Foreign Key(s): incident_id (references Incident_Report), hazard_id (references Hazards)

34.3. Attribute(s): None

34.4. Description: Links incidents to specific hazards present.

35. Priority_Level

35.1. Primary Key(s): priority_id

35.2. Foreign Key(s): None

35.3. Attribute(s): name

35.4. Description: Defines incident urgency (Low, Medium, High, Critical).

36. Saved_Station

36.1. Primary Key(s): saved_id

36.2. Foreign Key(s): user_id (references User), station_id (references Station)

36.3. Attribute(s): created_timestamp

36.4. Unique Constraints: (user_id, station_id) unique

36.5. Description: User bookmarks or favorite stations for quick access.

37. Upvote

37.1. Primary Key(s): upvote_id

37.2. Foreign Key(s): user_id (references User), incident_id (references Incident_Report)

37.3. Attribute(s): created_timestamp

37.4. Unique Constraints: (user_id, incident_id) unique

37.5. Description: Community validation mechanism allowing users to signal agreement with reported incidents.

38. Notification_Preference

38.1. Primary Key(s): pref_id

38.2. Foreign Key(s): user_id (references User)

38.3. Attribute(s): notification_type, enabled_flag

38.4. Unique Constraints: (user_id, notification_type) unique

38.5. Description: User-configurable settings for which alert types they wish to receive.

39. Notification

39.1. Primary Key(s): notification_id

39.2. Foreign Key(s): user_id (references User), incident_id (references Incident_Report, optional), line_id (references Line, optional)

39.3. Attribute(s): created_timestamp, delivery_status, notification_type

39.4. Description: Records of alerts sent to users about incidents, delays, or updates.

40. Trip

40.1. Primary Key(s): trip_id

40.2. Foreign Key(s): train_id (references Train), line_id (references Line)

40.3. Attribute(s): scheduled_start, scheduled_end

40.4. Description: Scheduled journey of a specific train along a specific line at a specific time.

41. Stop

41.1. Primary Key(s): stop_id

41.2. Foreign Key(s): trip_id (references Trip), station_id (references Station)

41.3. Attribute(s): scheduled_arrival, scheduled_departure, sequence_number (unique with trip_id)

41.4. Unique Constraints: (trip_id, station_id) unique, (trip_id, sequence_number) unique

41.5. Description: Individual scheduled station visits within a trip.

42. Live_Feed

42.1. Primary Key(s): feed_id

42.2. Foreign Key(s): stop_id (references Stop)

42.3. Attribute(s): update_timestamp, estimated_arrival_time, capacity_percentage

42.4. Description: Real-time updates to scheduled stops with current estimated arrival times and train capacity.

43. Delay_Propagation

43.1. Primary Key(s): propagation_id

43.2. Foreign Key(s): incident_id (references Incident_Report), trip_id (references Trip), stop_id (references Stop)

43.3. Attribute(s): delay_minutes

43.4. Description: Tracks how a single incident causes cascading delays downstream. Records calculated impact on subsequent stops.

44. Update

44.1. Primary Key(s): update_id

44.2. Foreign Key(s): incident_id (references Incident_Report), user_id (references User)

44.3. Attribute(s): comment_text, created_timestamp

44.4. Description: Public comments and status updates posted by users about incidents for community transparency.

45. Vote

45.1. Primary Key(s): vote_id

45.2. Foreign Key(s): user_id (references User), update_id (references Update)

45.3. Attribute(s): vote_type

45.4. Unique Constraints: (user_id, update_id) unique

45.5. Description: Community feedback on public updates to help surface accurate information.

46. Responder

46.1. Primary Key(s): responder_id

46.2. Foreign Key(s): None

46.3. Attribute(s): name (unique), description

46.4. Description: Lookup table for first responder types (Police, Fire, EMS) for proper dispatch categorization.

47. Dispatch_High_Level

47.1. Primary Key(s): high_level_dispatch_id

47.2. Foreign Key(s): incident_id (references Incident_Report), responder_id (references Responder), station_id (references Station)

47.3. Attribute(s): dispatched_timestamp

47.4. Description: Records of first responder units sent to incidents at specific stations.

48. Fare_Log

48.1. Primary Key(s): log_id

48.2. Foreign Key(s): incident_id (references Incident_Report), user_id (references User), station_id (references Station)

48.3. Attribute(s): detected_timestamp

48.4. Description: Automated records of fare evasion detection linked to low-priority incidents.

49. Data_Feed

49.1. Primary Key(s): feed_id

49.2. Foreign Key(s): None

49.3. Attribute(s): feed_name (unique), feed_type

49.4. Description: Metadata about external data sources integrated with the system (e.g., weather APIs, schedule feeds)